

Monalisa Burma

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PROFESSIONAL SUMMARY

Data Scientist with 1+ years of experience building and deploying production-grade ML and GenAI solutions. Specialized in LLM applications, NLP, and scalable AI systems using Python, FastAPI, and cloud platforms.

TECHNICAL SKILLS

Languages: Python, SQL

Tools: Power BI, Tableau, Docker, Git, Google Colab

Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, LSTM, XGBoost, LightGBM, Transformers, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Plotly

AI / ML: Supervised & Unsupervised Learning, Time Series Forecasting, Feature Engineering, Transfer Learning, Hyperparameter Tuning, Class Imbalance Handling, Model Evaluation & Validation

GenAI & LLMs: GPT Models, LangChain, LangGraph, Hugging Face, Prompt Engineering, RAG, Vector Databases, Azure OpenAI, Agentic AI, Embeddings, Fine-tuning

NLP & CV: NER, Text Summarization, Semantic Search, Token Classification, Image Classification, Document Parsing, Sentiment Analysis

Web & MLOps: FastAPI, Flask, Streamlit, n8n, MLflow, GitHub Actions, CI/CD

EXPERIENCE

Machine Learning Engineer / Data Scientist

May 2026 – Present

Portalwiz Technologies Pvt. Ltd.

Pune, India

- Performed data analysis and preprocessing on industrial sensor time-series data from Virtuoso client's plant, analyzing sensor behavior during plant stability conditions to identify alarm patterns and early warning signals.
- Built an LSTM-based deep learning model to predict equipment alarm conditions up to 5–10 minutes in advance from real-time industrial sensor data, achieving 100% detection rate.
- Designed a custom loss function to handle rare-event class imbalance, improving early-warning performance so that 80% of alarms were flagged at or before they occurred, enabling proactive operator response instead of reactive alerts.
- Engineered and selected predictive features from multi-sensor time-series data (trend, rate-of-change, rolling statistics), running comparative experiments to optimize model accuracy.
- Conducted root-cause analysis on model limitations and delivered findings to stakeholders through clear technical documentation, translating machine learning results into actionable business insights.

Data Scientist

Sep. 2024 – May 2026

BI Hub Solutions

Remote

- Led development of an AI-driven Tableau-to-Power BI migration tool, converting dashboards, KPIs, parameters, and calculations with 100% metadata retention.
- Built an LLM + NLP pipeline using FastAPI to transform .twb/.twbx files into .pbip with accurate DAX and visual mappings.
- Designed and launched an AI-powered DAX optimizer using OpenAI GPT and fine-tuned LLaMA, integrated with Azure Analysis Services.
- Reduced manual migration effort by 90% and improved dashboard load times by 70% and model analysis speed by 60%.

Data Science Intern

Jan 2024 – Feb 2024

NullClass

Remote

- Implemented a CNN-based eye-state classifier integrated with facial landmark tracking for live video streams.
- Achieved 95%+ accuracy and implemented real-time alert mechanisms to improve road safety.
- Validated the system for deployment in driver monitoring and fatigue prevention scenarios.

Machine Learning Intern	Dec 2023 – Jan 2024
<i>Feynn Labs</i>	<i>Remote, India</i>
- Analyzed consumer behavior in the EV market using real-world datasets including Indian EV market and McDonald’s customer data. - Applied unsupervised and supervised ML techniques including clustering and regression to identify consumer segments and behavioral trends.	

PROJECTS

Enhanced Travel Planner <i>FastAPI, Google Gemini, REST APIs</i>	
- Built a full-stack GenAI travel planner using FastAPI, React, and Google Gemini to generate personalized itineraries. - Integrated real-time weather, currency conversion, interactive maps, budget planning, and PDF export via REST APIs. - Integrated an AI-powered conversational chatbot with integrated voice assistant capabilities for real-time travel guidance and itinerary customization. - Designed a scalable backend–frontend architecture with environment-based config, CORS handling, and Docker.	
Stock Price Forecasting Application <i>LSTM, Python, Streamlit</i>	
- Developed an LSTM-based time-series forecasting system using 20 years of AAPL stock data. - Achieved 94% test accuracy with RMSE of 2.31 through automated preprocessing and feature engineering. - Deployed an interactive Streamlit dashboard with technical indicators for real-time analysis.	

EDUCATION

Indira Gandhi Institute of Technology	2019 – 2023
<i>Bachelor of Technology in Mechanical Engineering CGPA: 8.79</i>	<i>Odisha, IN</i>

CERTIFICATIONS

Programming Essentials in Python <i>Cisco Networking Academy</i>	Jun 2021
Gained strong foundations in Python programming, including data types, control flow, functions, and basic problem-solving techniques.	
Classify Images with TensorFlow on Google Cloud <i>Google</i>	May 2024
Learned to build, train, and deploy image classification models using TensorFlow on Google Cloud infrastructure.	
Cloud SQL <i>Google</i>	May 2024
Worked with managed relational databases on Google Cloud, including SQL-based data storage, querying, and performance optimization.	